

JAVIER REBULL, M.Sc.

Senior Full Stack Engineer | AI & ML Specialist | Software Architect
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Summary

Senior Full Stack Engineer and AI/ML Specialist with 20 years of experience delivering mission-critical technology for regulated financial services, including banking, payments, credit cards, and enterprise platforms, from discovery and technical scoping through system design, build, and production rollout. Hands-on in deploying machine learning models to production, building MLOps pipelines, and leading large-scale platform migrations without critical incidents, aligning executives, engineering teams, risk partners, and external banks around measurable outcomes.

Technical Skills

Programming Languages: Python, Java, JavaScript, TypeScript, SQL, COBOL, C/C++, C#, SAS
AI / Machine Learning: Deep Learning, NLP, Time Series Forecasting, Computer Vision, Generative AI, LLMs, MLOps
Frameworks & Libraries: PyTorch, Scikit-learn, XGBoost, Pandas, NumPy, Prophet, DeepAR, React, Angular
Cloud & DevOps: AWS (SageMaker), Mainframe-to-Cloud Migration, CI/CD, Docker, Linux, Git/GitHub
Software Engineering: Full Stack, System Architecture, Distributed Systems, Microservices, REST/SOAP, Agile/Scrum
Databases: Oracle, DB2, MySQL, MongoDB
Financial Technology: Core Banking, Payments, Credit Cards, Loans, Regulatory (FDIC 370), High-Availability Systems

Work Experience

Sr. Associate Application Development Sep 2019 – Present

Santander Bank US (Associate Application Development, Sep 2019 – Feb 2024; Sr. Associate since Mar 2024) *Quincy, MA*

- Led the Gravity Middleware Project end to end, migrating IBM Mainframe/DB2 to private cloud (Microfocus/Oracle) with zero critical incidents, and moved 500+ WAS/WebSphere servers to modern cloud infrastructure.
- Built end-to-end MLOps pipelines for automated model deployment and monitoring in production banking environments.
- Directed full stack development of Certificates of Deposit for OpenBank, from front-end interfaces to back-end services.
- Delivered deadline-driven regulatory initiatives: SBNA Loans COVID-19 Relief Program and Regulatory FDIC 370.
- Designed UI/UX for internal banking applications, raising user satisfaction by 35%.
- Kept distributed banking systems incident-free through root-cause analysis across hardware, network, and service layers, and led design and code reviews on testability and code quality.

Computer Systems Analyst Apr 2013 – Sep 2019

Vass IT US *Cambridge, MA*

- Built Computer Vision solutions for document processing and fraud detection, integrating ML models into client systems.
- Led full stack development of loan-management applications and managed IT project portfolios and PMO activities.
- Owned software products from requirements through deployment and maintenance for client business goals.

Spec Analytics Analyst 2 Jun 2010 – Apr 2013

Citibank Banamex *Mexico City, Mexico*

- Designed and optimized secure distributed systems for global financial services, processing millions of daily transactions with high availability.
- Built data-driven algorithms (SAS/Python) that raised marketing response rates 20% and cut delinquency 10%.
- Built interactive dashboards and front-end interfaces for executive reporting and real-time data visualization.

Sr. Programmer Aug 2007 – May 2010

Accenture *Mexico City, Mexico*

- Developed and maintained large-scale core financial systems (COBOL, Java) for client banks BBVA Bancomer and Santander México, with zero critical incidents.
- Designed front-end user interfaces for mainframe applications, improving operational efficiency and developer experience.

Systems Analyst & IT Help Desk Analyst

Microsistemas — Ford Credit Company

Aug 2006 – Aug 2007

Mexico City, Mexico

- Help desk consultant for Ford's department: system stability, security, and software and hardware support.

Education

M.S. in Artificial Intelligence and Data Analytics

Universidad Autónoma de Ciudad Juárez (UACJ)

2025 – 2027 (Expected)

Ciudad Juárez, Mexico

Focus: Smart Optimization, Linear Optimization, Non-Linear Optimization, Data Engineering

M.Sc. in Applied Artificial Intelligence

Tecnológico de Monterrey (ITESM)

2024 – 2026 • Degree conferred

Mexico City, Mexico

GPA: 4.0/4.0 • Coursework: Data Science, Machine Learning, AI, Data Visualization, Fintech & IoT

B.S. in Computer Science & Electronics Engineering

The National Polytechnic Institute of Mexico (ESCOM IPN)

2002 – 2006

Mexico City, Mexico

GPA: 3.28/4.0 • Coursework: Programming, Software Engineering, AI, Networks, Data Structures, Algorithms & OS

AI / ML Projects

- **EpiForecast-MX — Epidemiological Forecasting Platform** (2025–2026): M.Sc. capstone at Tecnológico de Monterrey with the IMSS. Forecasts weekly notified cases for Mexico's 32 federal entities 52 weeks ahead, choosing per series among Prophet, DeepAR, Ensemble and Stacking through an auditable rule. In production.
- **Text Classification with RNNs — AG NEWS Dataset** (Mar 2025): Built an RNN in PyTorch to classify AG NEWS articles, covering data preprocessing, model architecture design, training pipelines, and evaluation.
- **IoT for Sustainable Urban Development** (Feb 2025): Investigated IoT solutions for traffic congestion in Boston and public lighting efficiency in Mexico City using distributed sensor networks.
- **Multilayer Fully Connected Network in NumPy** (Jan 2025): Implemented a neural network from scratch for ASL letter recognition, including forward/backward propagation and performance benchmarking.
- **Stock Price Prediction for Apple** (Nov 2024): Developed and compared LSTM, ARIMA, Prophet, and Naïve predictive models on historical market data using advanced statistical methods.

Conferences & Publications

- **ePoster — 26th WPA World Congress of Psychiatry**, Stockholm, Sweden (Sep 23–26, 2026), Digital Psychiatry track: *“Population-level prediction of depression, Alzheimer's disease, and Parkinson's disease incidence using artificial intelligence: implications for health service planning.”*
- **Oral presentation — CALASS 2026, Congrès de l'ALASS**, Université de Montréal, Canada (Aug 27–29, 2026): *“De la surveillance épidémiologique à l'intelligence prédictive : le rôle de l'intelligence artificielle dans l'organisation prospective des systèmes publics de santé.”* Presented EpiForecast-MX (Tecnológico de Monterrey & IMSS).
- **Poster — 2nd Public Health Conference & 2nd HSG Europe Preconference**, Skopje, North Macedonia (May 21–22, 2026): *“AI-Based Forecasting to Shift Epidemiological Surveillance toward Predictive Approaches: Depression, Alzheimer's, and Parkinson's as Use Cases.”*
- **Forecasting Dengue in Mexico: An End-to-End Data Science Story** — Medium. End-to-end data science article covering the full workflow for forecasting dengue cases in Mexico.

Certificates: rebull.org/constancias

Training & Certifications

Python Essentials 1 & 2 — Cisco (Jul 2026) • Lifelong Learning — Certiprof (Jan 2026) • AWS Certified AI Practitioner (Dec 2025) • Scrum Master Professional — SMP[®]C (Jun 2025) • Generative AI in Action — IBM SkillsBuild (Jun 2025) • AWS Certified Cloud Practitioner (Jan 2025)